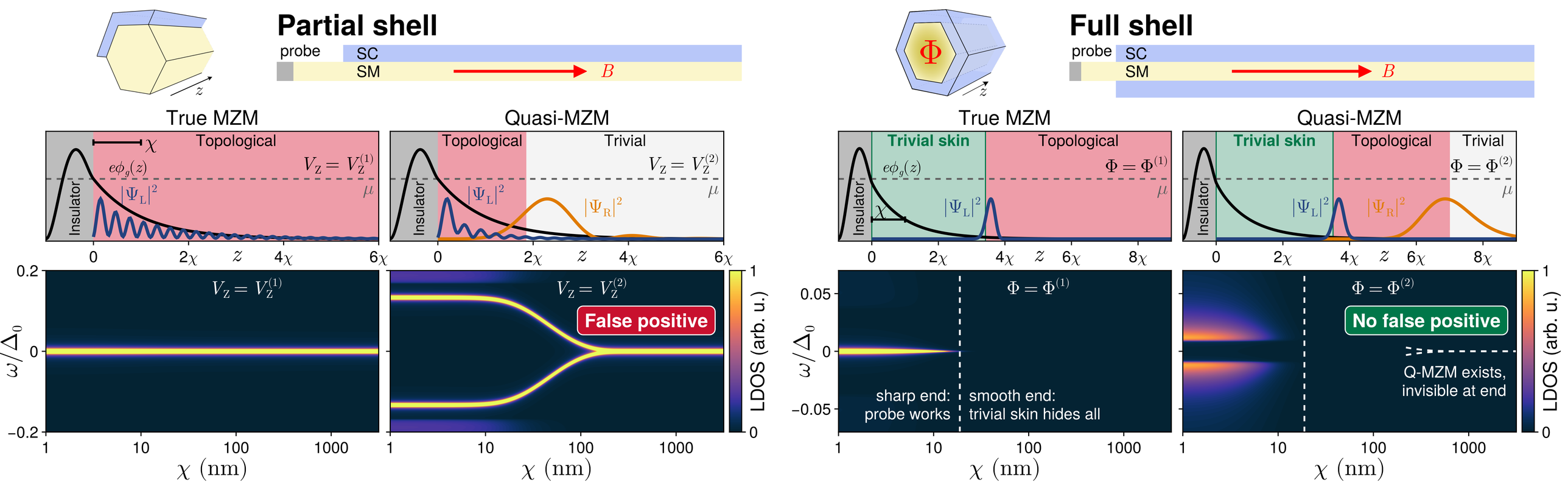

Partial shell
probe
SC
SM
B
z
True MZM
Quasi-MZM
Topological
Trivial
Insulator
χ
$e\phi_g(z)$
$V_Z = V_Z^{(1)}$
$V_Z = V_Z^{(2)}$
μ
$|\Psi_L|^2$
$|\Psi_R|^2$
0
2χ
4χ
6χ
z
ω/Δ_0
0.2
0
−0.2
False positive
1
10
100
1000
χ (nm)
LDOS (arb. u.)
1
0
Full shell
probe
SC
SM
B
Φ
z
True MZM
Quasi-MZM
Trivial skin
Topological
Trivial
Insulator
χ
$e\phi_g(z)$
$\Phi = \Phi^{(1)}$
$\Phi = \Phi^{(2)}$
μ
$|\Psi_L|^2$
$|\Psi_R|^2$
0
2χ
4χ
6χ
8χ
z
ω/Δ_0
0.05
0
−0.05
sharp end:
probe works
smooth end:
trivial skin hides all
No false positive
Q-MZM exists,
invisible at end
1
10
100
1000
χ (nm)
LDOS (arb. u.)
1
0